Hootan Seifi

AI Engineer | LLM Applications & Agentic AI Systems

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Portfolio: hootanseifi.pro

PROFESSIONAL SUMMARY

AI Engineer focused on designing, developing, evaluating, and deploying LLM-powered applications, Retrieval-Augmented Generation (RAG) systems, and AI agents. Hands-on experience building agentic workflows with LangGraph, implementing hybrid information retrieval, evaluating LLM outputs, and developing security and reliability mechanisms for AI applications.

Built practical projects including an enterprise AI agent platform, a bilingual RAG system, and a multi-agent Bitcoin trading simulator. Holds an M.Sc. in Artificial Intelligence, with previous experience in R&D, software application analysis, and financial-market analysis.

TECHNICAL SKILLS

LLM & Agent Engineering: Python, LangGraph, LangChain, OpenAI Agents SDK, CrewAI, MCP, Tool Calling, Prompt Engineering, Structured Outputs, LLM Evaluation, Guardrails

RAG & Information Retrieval: Embeddings, Chunking, pgvector, PostgreSQL, Chroma, Hybrid Search, BM25, Reranking, RRF, Recall@k, MRR

Backend & Testing: FastAPI, SQLAlchemy, Alembic, REST APIs, SSE, pytest, Playwright

Machine Learning & Data: PyTorch, scikit-learn, XGBoost, Pandas, SQL, QLoRA, PEFT

Infrastructure & Deployment: Docker, Docker Compose, Linux, Caddy, Git, GitHub Actions, CI/CD

Automation & BI: Workflow design with n8n, Power BI, SQL

Frontend: Next.js, TypeScript

Familiar with: AWS Bedrock, SageMaker, Lambda, Terraform, Modal, Vercel, Spark, Kafka, Airflow

SELECTED PROJECTS

hootanseifi.pro | AI Engineering Portfolio

- Built a bilingual RAG system that retrieves relevant information and generates source-grounded answers.
- Implemented hybrid retrieval by combining BM25 and pgvector, with Reciprocal Rank Fusion (RRF).
- Evaluated 72 scenarios covering citation quality, grounding, refusal behavior, and prompt-injection resistance; passed all 72 scenarios.
- Developed a seven-agent Bitcoin trading simulator using LangGraph and live market prices.
- Implemented independent risk-management rules for the AI agent, including a maximum 5x leverage, 20% margin requirement, and 1–5% stop-loss range. Users can select their own trading settings, while the AI agent continues to enforce its independent risk constraints.
- Completed 20 simulation sessions without violations of the AI agent's defined risk rules.
- Built a job-fit scoring system with programmatic score-range enforcement.
- Implemented daily and monthly AI usage budgets.
- Deployed a five-service architecture using Next.js, FastAPI, a worker, PostgreSQL, and Caddy on a VPS.
- Set up GitHub Actions CI/CD, automated tests, secret scanning, and a tested rollback mechanism.

Astra Enterprise AI Agent Platform

- Designed and developed an enterprise AI agent platform using FastAPI and LangGraph.
- Implemented dynamic routing across direct responses, RAG retrieval, tool calls, and combined RAG-plus-tool workflows.
- Built multi-step agent workflows with recovery nodes, PostgreSQL checkpointing, and human-in-the-loop approval.
- Developed a PostgreSQL/pgvector-based RAG pipeline with hybrid retrieval, HNSW indexing, reranking, and permission-aware retrieval.
- Improved Recall@1 from 0.67 to 1.00 on the project's evaluation dataset.
- Implemented RBAC/JWT authorization, tool-input validation, and Persian/English prompt-injection detection.
- Added secret redaction, rate limiting, Prometheus monitoring, and token/cost tracking.
- Developed an MCP server to expose tools through a standardized interface.
- Achieved 407 automated tests and 93% code coverage, with CI quality gates.

Price Estimate | Product Price Prediction

- Built a product price estimation system using a dataset of approximately 800,000 Amazon products.
- Evaluated models ranging from baseline approaches to deep residual neural networks.
- Fine-tuned Llama 3.2 3B using QLoRA, Hugging Face TRL, and Modal, and integrated the fine-tuned model into the system.
- Implemented a Chroma-based RAG pipeline to incorporate product information into price estimation.
- Used Non-Negative Least Squares (NNLS) for ensemble weight optimization and reweighting when model outputs were unavailable.
- Evaluated model performance using MAE, RMSLE, R^2, and Hit Rate, including 95% confidence intervals for MAE.
- Developed deal-alert functionality and exposed the application through FastAPI and Gradio.
- Added more than 60 automated tests.

AI Medical Notes | Medical Note Assistant

- Built an application that transforms clinical visit notes into structured summaries, next steps, and draft patient emails.
- Implemented streaming responses using Server-Sent Events (SSE).
- Developed the backend with FastAPI and the frontend with Next.js.
- Integrated the OpenAI API and Clerk for AI capabilities and authentication.
- Deployed the application on Vercel.

Behengam | Integrated Healthcare Platform

- Contributing to the design and preparation of a healthcare platform intended to connect 12 stakeholder groups.
- Planning a shared health-record architecture and data exchange based on HL7 v2, FHIR, LOINC, and UCUM standards.
- Selected Django REST Framework and Kafka as part of the planned technical architecture.
- Defining AI-assisted alerting and decision-support use cases while keeping healthcare professionals responsible for final decisions.
- Currently in the preparation and design phase.

PROFESSIONAL EXPERIENCE

Independent AI Engineer Aug 2026 – Present
Design, build, test and deploy LLM and agentic applications end to end. All code is public on GitHub.

AI Product Development Manager — Eghtesad Gostar Behsaman May 2025 – Apr 2026
Responsible for developing AI-based products, as well as their marketing and overall growth.

Senior Power Plant Specialist — Taban Energy Power Plant Mar 2025 – May 2025
Technical responsibilities and oversight of the performance of plant equipment and operational processes.

Fundamental Analyst & Trader — Novin Eghtesad Technologies Sep 2022 – Feb 2025
Analyzed financial and crypto markets, evaluated blockchain projects for analytical reports, and developed a Python trading bot on market data.

Technical Specialist — Cafe Bazaar Jan 2022 – May 2022
Technical review of applications, evaluation of their performance quality, and collaboration in the review and publishing process.

R&D Specialist — MECO / MAPNA Apr 2020 – Mar 2021
Contributed to research and development activities and technical review of projects in the company's field.

EDUCATION

M.Sc. in Electrical Engineering – Artificial Intelligence — Amirkabir University of Technology, Tafresh Branch 2016–2018

B.Sc. in Electrical Engineering – Electronics 2011–2015

ACADEMIC PROJECTS

- Ethereum transaction fraud detection using XGBoost.
- Sentiment analysis of tweets and its relationship with Nvidia stock prices using NLP and time-series methods.
- Traffic sign classification using CNNs and the GTSRB dataset.

CERTIFICATIONS

- LLM Engineering — Ed Donner
- Machine Learning Specialization — Andrew Ng
- Financial Trading in Python — DataCamp

LANGUAGES

- Persian: Native
- English: Professional Working Proficiency